Coulomb excitation 2023Dr01

2023Dr01: An intermediate Coulomb excitation experiment was performed to measure the B(E2, $0^+_1 \rightarrow 2^+_1$) value of ^{36,38}Ca.
^{34}Ar beam was a contaminant produced from the fragmentation of a 140 MeV/nucleon ^{40}Ca primary beam delivered by the Coupled Cyclotron Facility at the National Superconducting Cyclotron Laboratory (NSCL) impinging upon an 800 mg/cm^2 thick ^{9}Be production target and separated using a 300 mg/cm^2 Al wedge degrader in the A1900 fragment separator. Secondary beam was scattered off a 257 mg/cm^{+2} thick Au foil located at the target position of the S800 spectrograph. The deexcited γ-rays were detected using GRETINA surrounded S800 spectrograph. The GRETINA consisted a total of 48 HPGe crystals; 32 of them at 90° and 16 of them at 58° with respect to the beam direction. The inelastic excitations of the projectile and target nuclei were quantified with the detection of prompt γ-rays in GRETINA. The scattered projectiles were identified using S800 focal plane detection system together with tof measurement. Measured: Eγ, tof, cross section and B(E2). Comparison with the shell model calculations.

^{34}Ar Levels

E(level)†	J$^{\pi}$‡	Comments
0.0	0^+	
2091 *2*	2^+	B(E2)↑=0.0293 *16* σ=52.1 mb *29* measured from laboratory scattering angles θ=0° to θ_{max}=55° mrad at 64.3 MeV/nucleon mid-target energy.

† From measured Eγ (2023Dr01).
‡ From the Adopted Levels.

γ(^{34}Ar)

E$_\gamma$	E$_i$(level)	J$^{\pi}_i$	E$_f$	J$^{\pi}_f$
2091 *2*	2091	2^+	0.0	0^+

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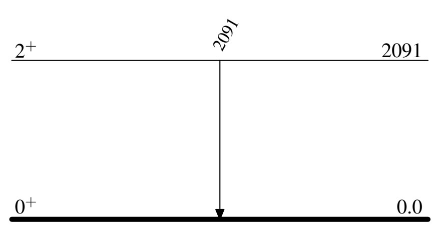
Level Scheme



$^{34}_{18}Ar_{16}$